

Andrew Bacigalupi

St. Paul, MN, 55116 • andrew.bacigalupi@gmail.com • 605-252-1868

Education

Tufts University

Medford, MA

- B.S. Computer Science (School of Engineering), B.S. Applied Mathematics, expected May 2028
- Pursuing Minor in Philosophy, Dean's List all semesters
- **Relevant Coursework:** Data Structures, Algorithms, Machine Learning, Machine Structure/Assembly, Probability, Discrete Math, Linear Algebra, Differential Equations

Highland Park Senior High School

St. Paul, MN

- Full International Baccalaureate Diploma, May 2024
- 4.86 GPA, Summa Cum Laude, High Honors in Mathematics and French

Experience

Teaching Assistant: Tufts Computer Science Department (CS15: Data Structures)

Jan 2025 – Present

- Provide constructive feedback on weekly student homework, labs, and projects in a timely manner
- Grade homework assignments weekly and exams when applicable and lead weekly office hour sessions for students, fostering a collaborative learning environment and enhancing problem-solving skills.

Volunteer Software Developer: [Tufts JumboCode](#)

Jan 2025 – Present

- Addressing disconnected communication issues for Commonwealth Kitchen, a local non-profit that supports small food businesses and food shelves.
- Leading a small team contributing to the development of a centralized communication app using Next.js, SQL, Tailwind CSS, and GitHub.

STEM Instructor: Lavner Education

Jun 2025 – Aug 2025

- Led classrooms of kids ages 6-14 on robotics engineering, computer science, chess, computer skills.
- Collaborated with fellow counselors and management to deliver fun, engaging, safe camp experiences.

Actuarial Analyst Intern: Securian Financial

Mar 2023 – May 2023

- Prepared audits by scheduling check-ins, revised and finalized quarterly report documents.
- Collaborated with other analysts to brainstorm improvements to risk models and data analysis

Skills

- **Languages:** Python, Java, C/C++, SQL, Assembly, Typescript, MATLAB
- **Data Science:** NumPy/SciPy, Scikit-Learn, Matplotlib, Notebook, Seaborn, Pandas, Excel, MediaPipe
- **Software Development:** React/Next.js, HTML/CSS/Tailwind, Software Design, Git/[GitHub](#)

Projects

[Iterly](#): Founding Engineer, Writer

- Entry-level learn-to-code site that emphasizes problem solving on large scale input data for hands-on and meaningful practice. Features educational articles/suggestions and code/puzzle contributions.
- Built with Next.js/TypeScript/Tailwind CSS, Neon/Drizzle for DB, and GitHub OAuth.

Tufts Men's Basketball Project Three-Point Margin: Software Engineer, Stat Tracker

- Designed/implemented a custom web app with Google Apps Script/Sheets for in-practice stat-tracking.
- Produced personnel insights and weighted statistics to aid in coaches' meetings and strategic decisions.

Golf Swing Analysis Application

- Combined Google's MediaPipe library with statistical analysis to quantify and analyze golf swings.
- Integrated the analysis with MySQL database to save results, accessible with graphical user interface.

Interests

- **Professional:** Computational Bio/Astro/Ecology, ML, Medical Tech, Economics, Public Health
- **Personal:** Basketball + Tennis, Running + Weightlifting, Piano + Singing, Space + Aliens, Chess